

Panel A: green = Same, orange = Different, purple = Opposite; (0,0) remains in the score denominator but is not tiled. Missing states are excluded; * denotes directional BH FDR ≤ 0.05 . Panel B uses categorical Yu-style score-tail/APOE positions; dot size = GeneRatio and color = $-\log_{10}(\text{BH-adjusted } P)$.